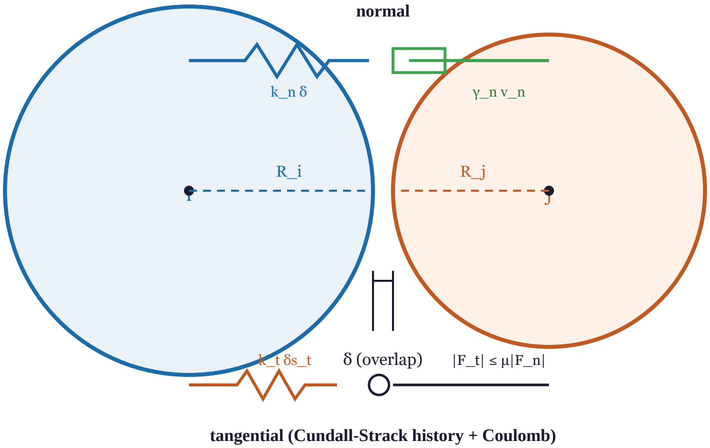
Soft-sphere contact: two overlapping particles



contact law

$F_n = k_n \delta - m^* \gamma_n v_n$

$F_t = -k_t \delta s_t - m^* \gamma_t v_t$

$|F_t| \leq \mu |F_n|$

Hertz: scale by

$\sqrt{\delta} \cdot \sqrt{(R_i R_j / (R_i+R_j))}$

damping from restitution e:

$\gamma_n = -2 \ln e \sqrt{(k_n/m^*)}$

$/ \sqrt{(\pi^2 + \ln^2 e)}$

$dt << t_c = \pi\sqrt{(m^*/k_n)}$

Forms verified verbatim against the LAMMPS granular pair-styles and the tumbling-mill DEM literature.